Artun Balta

Ankara, Turkey · relocating to London, UK from September 2026 · +90 530 460 97 19
artun.balta@outlook.com · github.com/artunbalta · linkedin.com/in/artunbalta

Profile

Electrical and electronics engineer working at the intersection of neural signal processing and applied machine learning. Incoming MRes Neurotechnology researcher at Imperial College London, decoding auditory attention from single-neuron recordings, and co-founder of a funded AI voice-agent company with a shipped product and five clinic pilots.

Education

Imperial College London · *MRes Neurotechnology (incoming)* London, UK · 2026 – 2027

- Thesis (supervisor: Dr. Andriy Kozlov): *Auditory Attention Decoding, Adapting Dynamic Spectrotemporal Receptive Fields to Single-Neuron Spike Recordings.* Extends deep-network dSTRF estimation from population-level ECoG to single-unit spike trains, introducing Poisson likelihood formulations and sparsity handling to characterise nonlinear tuning in the auditory pathway.

İhsan Doğramacı Bilkent University · *B.Sc. Electrical and Electronics Engineering* Ankara, Turkey · 2021 – 2026

- **1st Place** (12 teams), Graduation Project Competition · **2nd Place** (45 teams), Bilkentpreneurs Challenge 2026, $3,000 in prize funding · Bilkent Entrepreneurship Award.

Lycée Tevfik Fikret · *High School, Mathematics & Sciences* Ankara, Turkey · 2010 – 2021

- High Honour graduate on academic scholarship · Top 0.2% nationally in the YKS 2021 entrance exam among 3M+ candidates · **2nd Place**, Mathématiques Sans Frontières.

Research & Projects

Parallel MRI Reconstruction: Kernel Geometry under Anisotropic Sampling

- Gained 0.65 dB and matched, with 11 calibration taps, the reconstruction quality the standard rule needed 19 taps to reach, by deriving the kernel-support rule from acquisition physics rather than an arbitrary index-space convention. Validated across 8,947 paired reconstructions against an exact null control.
- Showed that a published kernel-shape equivalence breaks down on anisotropic acquisition grids, and exposed a fault in the reference pipeline that invalidated 27 of 36 cells in the comparison it was built to support.

Decoding Imagined Speech from Non-Invasive EEG

- Reached 46% cross-validated accuracy against a 25% chance baseline. Built a kernel SVM with sequential minimal optimisation, Bayesian logistic regression and an EEGNet-class convolutional network from first principles in NumPy, hand-deriving every gradient with no autograd. A cross-corpus control isolated the near-chance ceiling as a property of the dataset rather than the pipeline.

Autonomous Jenga-Playing Robot

- Hierarchical control stack coupling a learned discrete block-selection policy to an optimisation-based motion planner, bridging reinforcement learning and continuous trajectory synthesis for contact-rich manipulation.

Further coursework projects: medical visual question answering by low-rank adaptation of a vision-language model · convolutional classifier built from scratch to discriminate real from synthetic faces · FPGA colour-sensing joint controller in VHDL · ARM Cortex-M0+ interactive embedded system · Tesla coil and JFET high-impedance probe for non-contact electric field measurement.

Experience

Volina AI · volina.ai · *Co-Founder* Ankara, Turkey · 2025 – present

- AI voice-agent platform for Turkish health-tourism dental clinics, re-engaging dormant international leads with agents that profile callers in real time and adapt tone, objection handling and pacing mid-conversation.
- Delivered five clinic pilots covering roughly 40,000 leads, with per-pilot conversion rates of 1.2% to 3.8%. KVKK and GDPR-compliant call handling and consent capture built into the platform.

Boğaziçi University MIMLAB · *Volunteer Researcher* İstanbul, Turkey · 2025 – 2026

- Multimodal EEG preprocessing and classification: adaptive filtering against physiological reference channels, artefact evaluation, ERP extraction from n-back tasks, and machine-learning models for cognitive workload state classification.

ASELSAN · *Radar Systems Engineering Intern* Ankara, Turkey · July – August 2024

- Signal-processing optimisation for airborne radar systems: CFAR detection, pulse compression and pulse integration, improving detection accuracy and range through SNR enhancement.

Turkish Entrepreneurship Foundation (GirVak) · *Fellow* İstanbul, Turkey · 2023 – 2026

- One of 28 Fellows selected from 154,558 applicants to Türkiye's leading entrepreneurship programme; previously one of 100 Challengers selected from 149,766 applicants.

Bilkent University Physics Laboratory · *Teaching Assistant* Ankara, Turkey · 2023 – 2024

- Teaching assistant for PHYS102 (General Physics II) and laboratory tutor for PHYS101 (General Physics I).

Skills, Languages & Interests

Programming: Python (NumPy, SciPy, PyTorch, scikit-learn, MNE, pandas), MATLAB, C, C++, JavaScript/TypeScript, VHDL
Methods: Neural signal processing (EEG, ECoG, ERP, spike trains), Bayesian inference, Monte Carlo simulation, deep learning, parallel MRI reconstruction, LLM fine-tuning (LoRA) · **Tools:** Git, Docker, LTspice, LaTeX
Languages: Turkish (native) · English (IELTS Academic 7.5) · French (DELF B2) · **Interests:** Stand-up comedy, mechanical watches, quantitative horse-racing form modelling